

Stimulus Event Reconstruction

The raw fNIRS data streams recorded by the NIRSport2 system contained multiple stimulus channels generated automatically by the ACELETRA® software during each reading session. Because these channels encoded individual trial-level events rather than block-level epochs suitable for the general linear model (GLM), a custom MATLAB script (`contagem_nova.m`) was developed to reconstruct block-level stimulus events from the raw onset markers prior to first-level GLM estimation.

For each participant-session pair, the script identified two classes of stimulus channels within the raw data object: (i) reading activity channels, containing the onset markers of individual reading stimuli and their offsets, and (ii) rest channels, containing the onset markers of inter-block rest periods. The rest channel was identified automatically as the stimulus channel with the fewest recorded events, reflecting the smaller number of rest periods relative to reading trials within each session.

Block boundaries were defined by the onset times of the rest periods, such that each block was delimited by two consecutive rest markers. Within each block, all reading onset and offset events were isolated. The block-level epoch was then characterised by three parameters: (i) onset, defined as the timestamp of the first reading stimulus within the block; (ii) duration, defined as the difference between the timestamp of the last reading offset and the onset of the first reading stimulus; and (iii) amplitude, set to unity (1) for all blocks, yielding a standard boxcar regressor.

The expected block structure varied by session type. The calibration session (`sessao_num = 0`) comprised four blocks, whereas each of the nine training sessions (`sessao_num = 1–9`) comprised seven blocks. Each block contained five reading stimuli (`NstimPerBlock = 5`). Automated integrity checks were applied to every participant-session pair: the reconstructed block count was compared against the expected value, and the number of stimuli detected per block was verified against the expected count of five. Sessions or blocks failing these checks triggered a runtime warning and were flagged for manual inspection.

Following event reconstruction, the original multi-channel stimulus structure was discarded using the `nirs.modules.DiscardStims` module of the NIRS Toolbox, and a single compound stimulus object (labelled “Aceletra”) was created using `nirs.design.StimulusEvents`. This unified stimulus object, containing block-level onsets, durations, and unit amplitudes, was assigned to each data object and subsequently used as the sole task regressor in the first-level GLM estimation (see fNIRS Data Preprocessing).